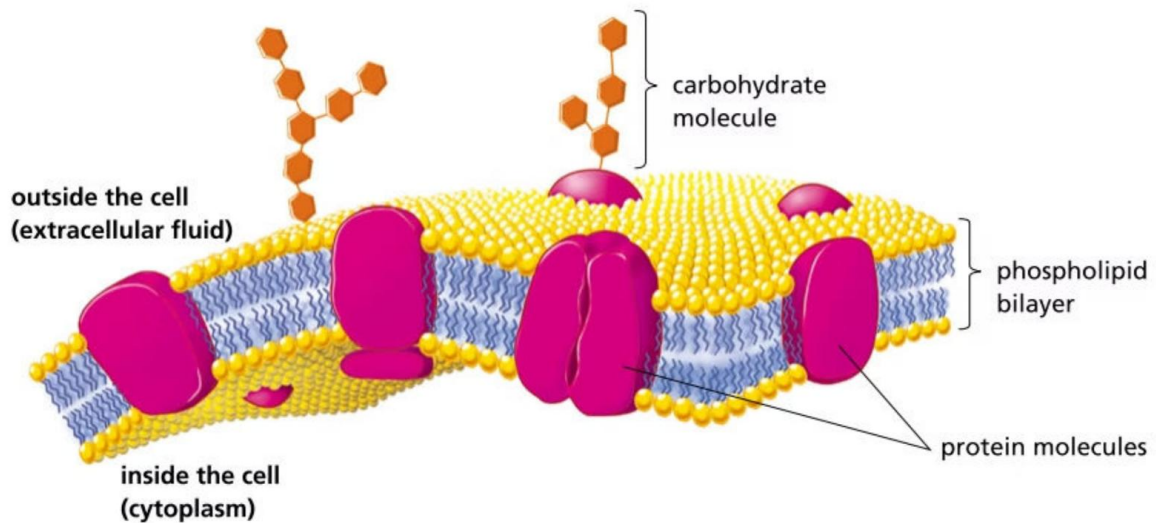
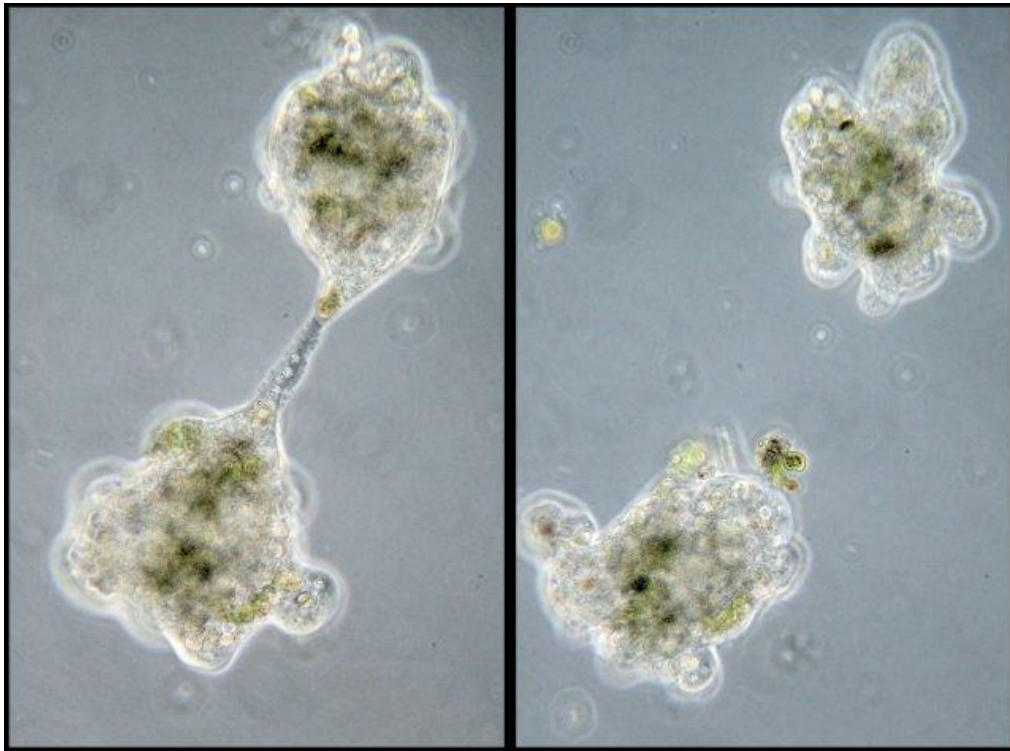


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Ch 3 Membrane transport – Cell membrane and Osmosis

Fluid Mosaic Model 流動鑲嵌模型 of cell membrane



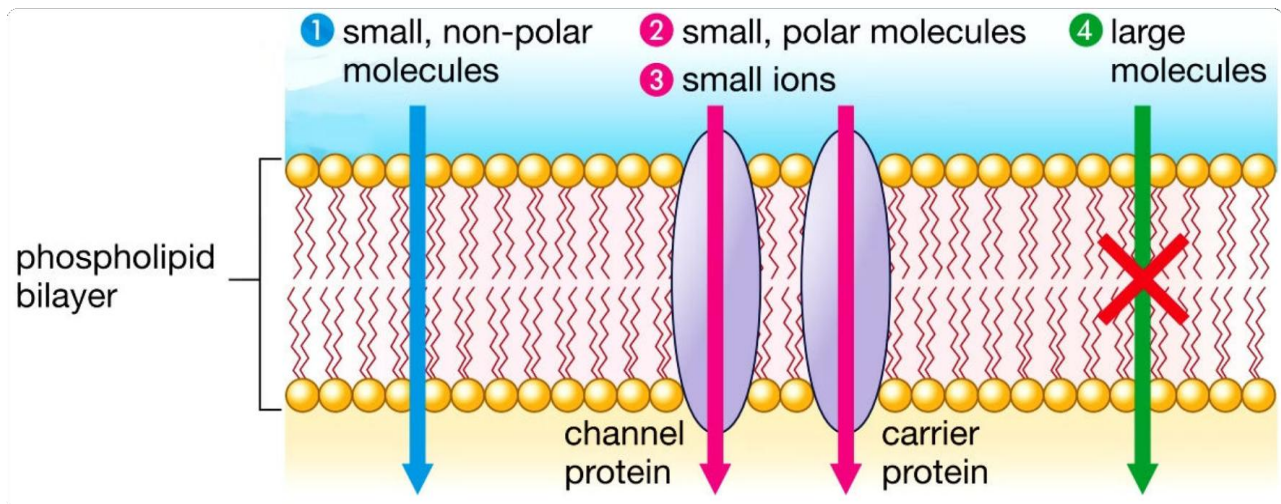
Fluid 流動	<u>Phospholipid bilayer</u> 磷脂雙層 is <u>fluid</u> and could move laterally 橫向移動 together with the embedded proteins, infer flexibility 靈活性:
Mosaic 馬賽克	<u>Membrane proteins</u> 膜蛋白 Embedded in a <u>mosaic pattern</u>

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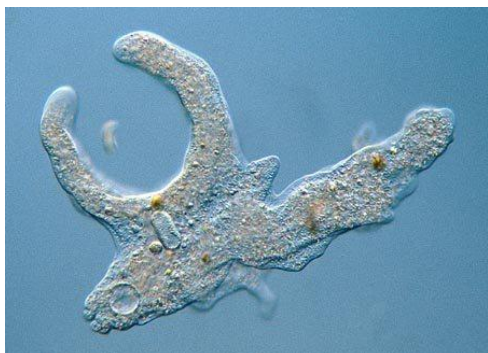
Differential permeability 差異滲透性 of cell membrane

似水: Polar 極性 molecules are water-soluble (hydrophilic 親水);

似油: Non-polar 非極性 molecules are lipid-soluble (hydrophobic 疏水)



Substance	Examples	Movement across membrane
Small, non-polar molecules 非極性小分子	Oxygen, carbon dioxide, Fatty acids 脂肪酸, glycerol 甘油	Phospholipid bilayer
Small, polar molecules 極性小分子	Water, amino acids 胺基酸, glucose 葡萄糖	Proteins
Small ions (with charge) 帶電荷的小離子	Sodium 鈉, calcium 鈣	Proteins
Large molecules 大分子	Starch, triglycerides 三酸甘油酯, proteins 蛋白質	Membrane folding



Membrane folding is also involved in eating by some unicellular organisms 單細胞生物.

Membrane folding is enabled by which property of the cell membrane?

"Fluid" 流動 / **"Mosaic"** 馬賽克

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Concept checks

Multiple Choice Questions (MCQs)

1. Which of the following is an example of a small, non-polar molecule that can diffuse through the phospholipid bilayer? (non-polar = "like oil")
 - A. Water
 - B. Oxygen
 - C. Glucose
 - D. Sodium ions

2. Which of the following molecules requires a protein channel to cross the cell membrane?
 - A. Carbon dioxide
 - B. Fatty acids
 - C. Sodium ions
 - D. Glycerol

3. What is the main function of the phospholipid bilayer in the cell membrane?
 - A. To provide energy for cellular activities
 - B. To act as a barrier controlling materials moving in and out of the cell
 - C. To synthesize proteins
 - D. To store genetic material

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4. Which of the following best describes the term "fluid" in the Fluid Mosaic Model?



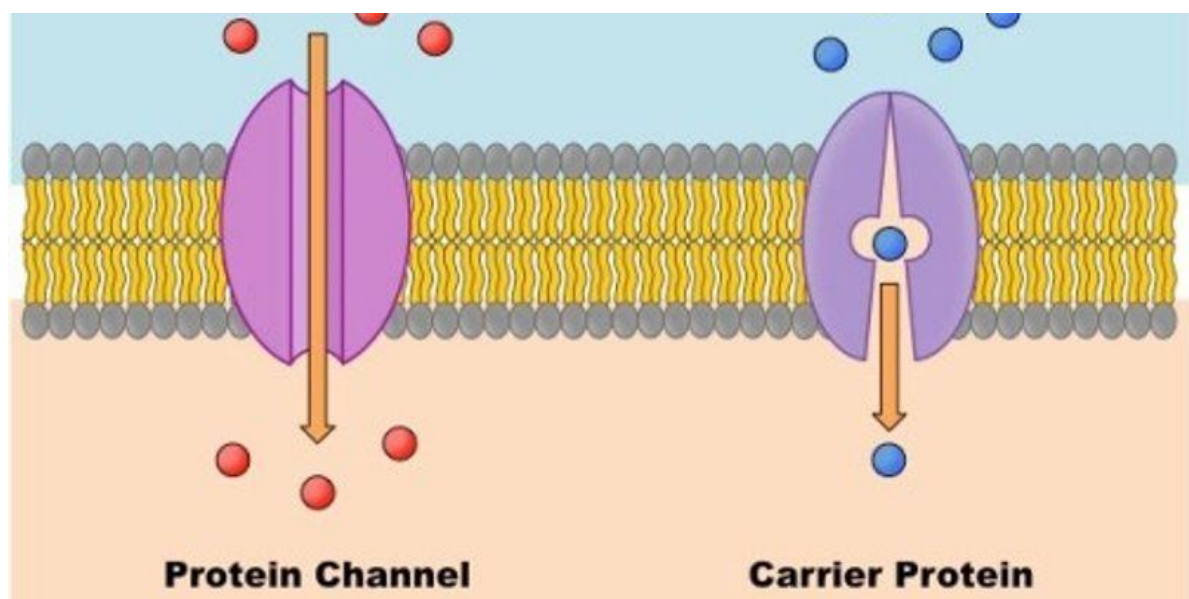
- A. Embedded proteins are rigid 硬.
 - B. Phospholipids can move laterally.
 - C. The cell membrane is impermeable 不透.
 - D. Membrane proteins are fixed in position.
5. What is the characteristic of non-polar molecules that allows them to pass through the phospholipid bilayer?
- A. They are large and hydrophilic 親水.
 - B. They are small and hydrophobic 疏水.
 - C. They are charged 帶電荷.
 - D. They are polar 極性 .

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6. Which of the following is NOT a property of the cell membrane?

- A. Differential permeability
- B. Flexibility
- C. Mosaic pattern of proteins
- D. Rigid and immovable structure

7. What is the role of proteins in the cell membrane?



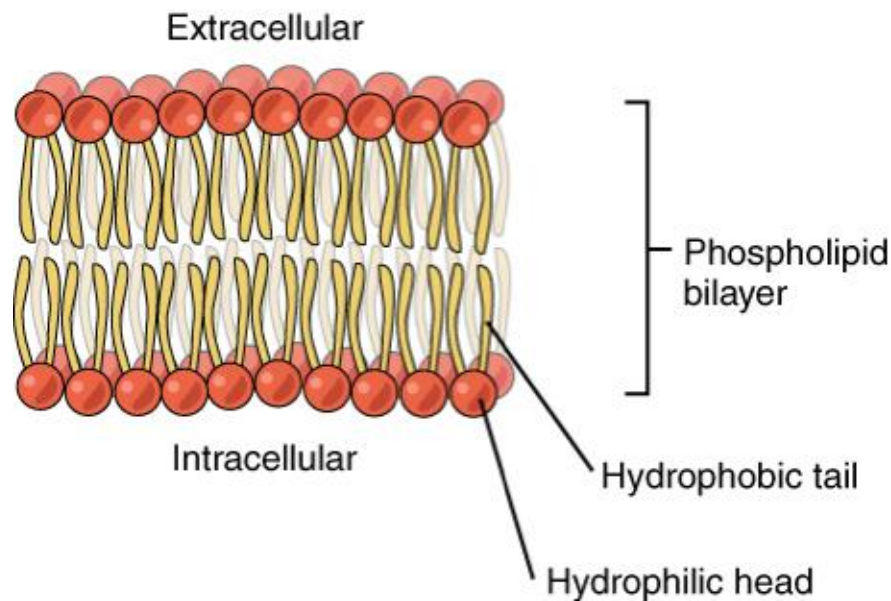
- A. To allow the passage of small, non-polar molecules
- B. To transport large or charged molecules across the membrane
- C. To synthesize carbohydrates
- D. To provide energy

8. Which of the following molecules is transported via membrane folding?

- A. Oxygen
- B. Water
- C. Starch
- D. Glucose

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9. Which part of the cell membrane is hydrophobic 疏水?



- A. The phosphate head of the phospholipids
- B. The protein channels
- C. The fatty acid tails of the phospholipids
- D. The carbohydrate molecules

10. What does the term "mosaic 馬賽克" in the Fluid Mosaic Model refer to?



- A. The flexibility of the phospholipid bilayer
- B. The arrangement of proteins embedded in the membrane
- C. The selective permeability of the membrane
- D. The movement of ions across the membrane

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True/False Questions

Directions: Put a "✓" if the sentence is correct and put a "X" on the phrase or words that are incorrect.

1. The cell membrane is made up of a rigid structure to provide support.
2. Small, non-polar molecules like oxygen can diffuse through the phospholipid bilayer.
3. Carrier proteins are responsible for transporting small, non-polar molecules.
4. The Fluid Mosaic Model describes the cell membrane as a flexible and dynamic structure.
5. Polar molecules can pass through the phospholipid bilayer without assistance.

Fill in the Blanks

Word Bank:

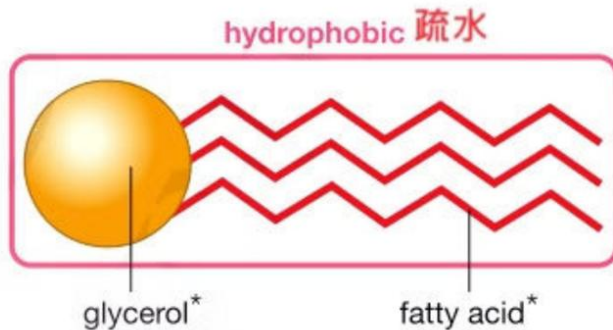
☐ phospholipid bilayer 磷脂雙層 ☐ hydrophilic 親水性 ☐ hydrophobic 疏水性
☐ proteins 蛋白質 ☐ differential permeability 差異滲透性

1. The _____ forms the basic structure of the cell membrane.
2. Molecules that are _____ interact with water and are polar.
3. Molecules that are _____ do not dissolve in water and are non-polar.
4. _____ are embedded in the cell membrane to aid in the transport of certain molecules.
5. The property of _____ allows the cell membrane to control what enters and exits the cell.

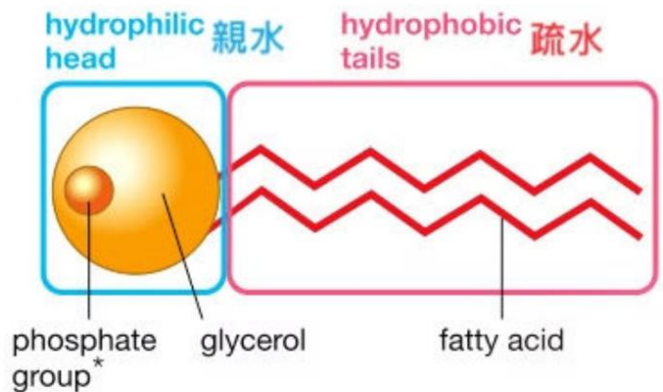
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Phospholipid 磷脂

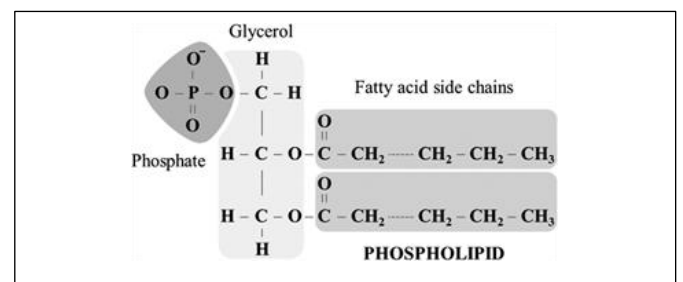
Triglyceride molecule



Phospholipid molecule

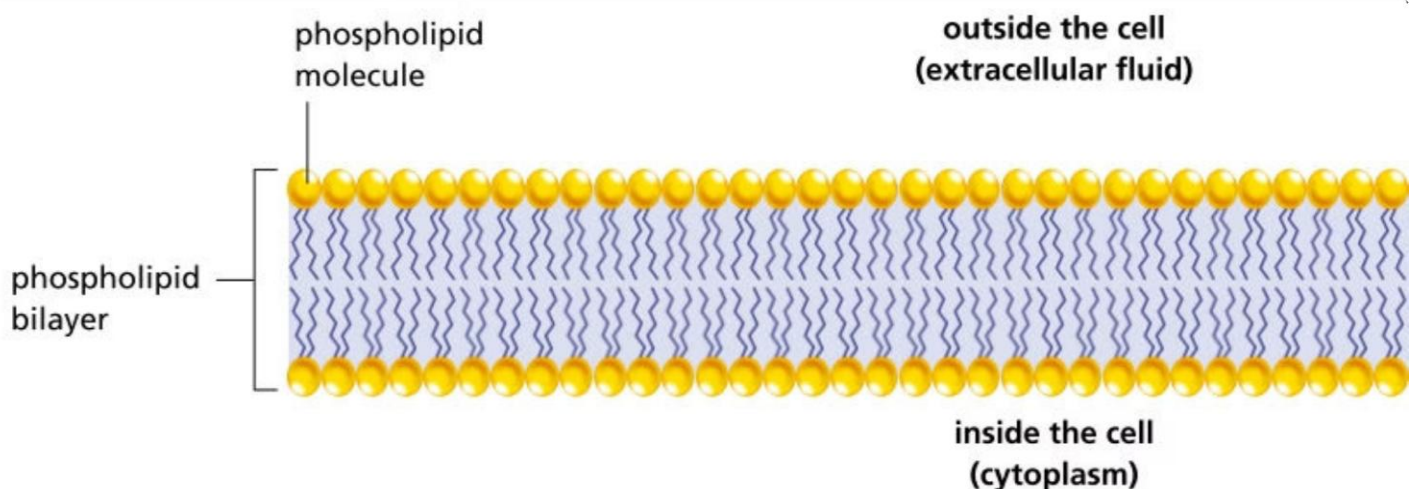


	Structure	Relationship with water
Triglycerides	1 glycerol + 3 fatty acids	Hydrophobic (non-polar)
Phospholipids	1 glycerol + 2 fatty acids + 1 phosphate group (polar)	Hydrophilic head (polar) Hydrophobic tail (non-polar)



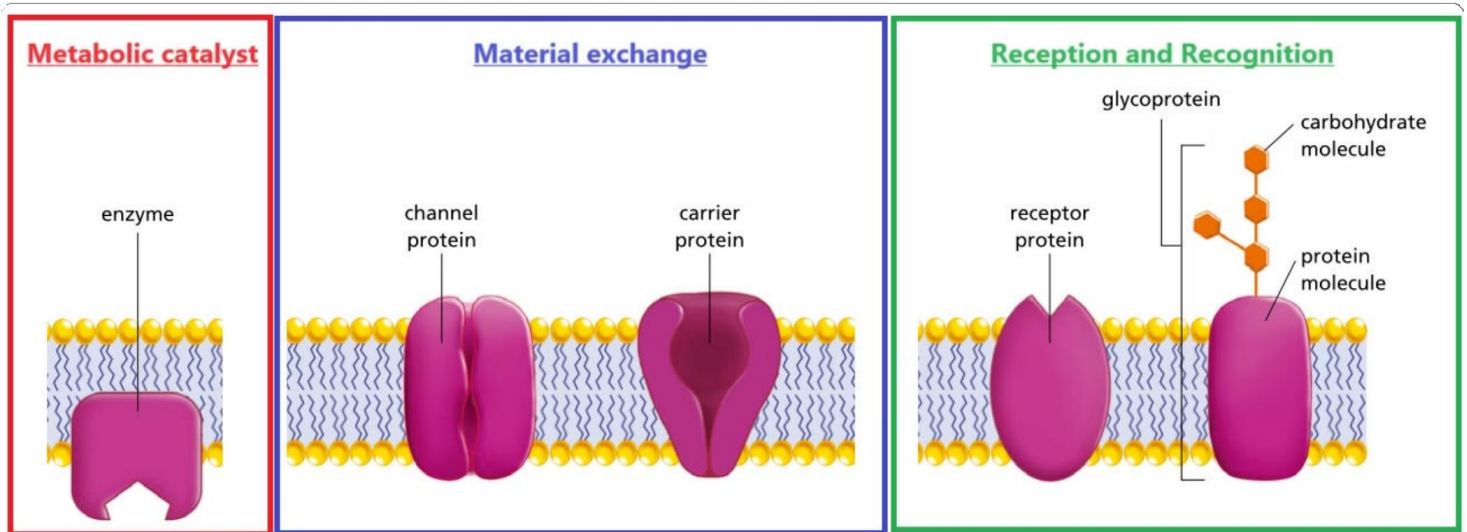
Phospholipid bilayer 磷脂雙層

hydrophilic 親水 head + hydrophobic 疏水 tail => **"Tail-to-tail" bilayer**



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Membrane proteins 膜蛋白

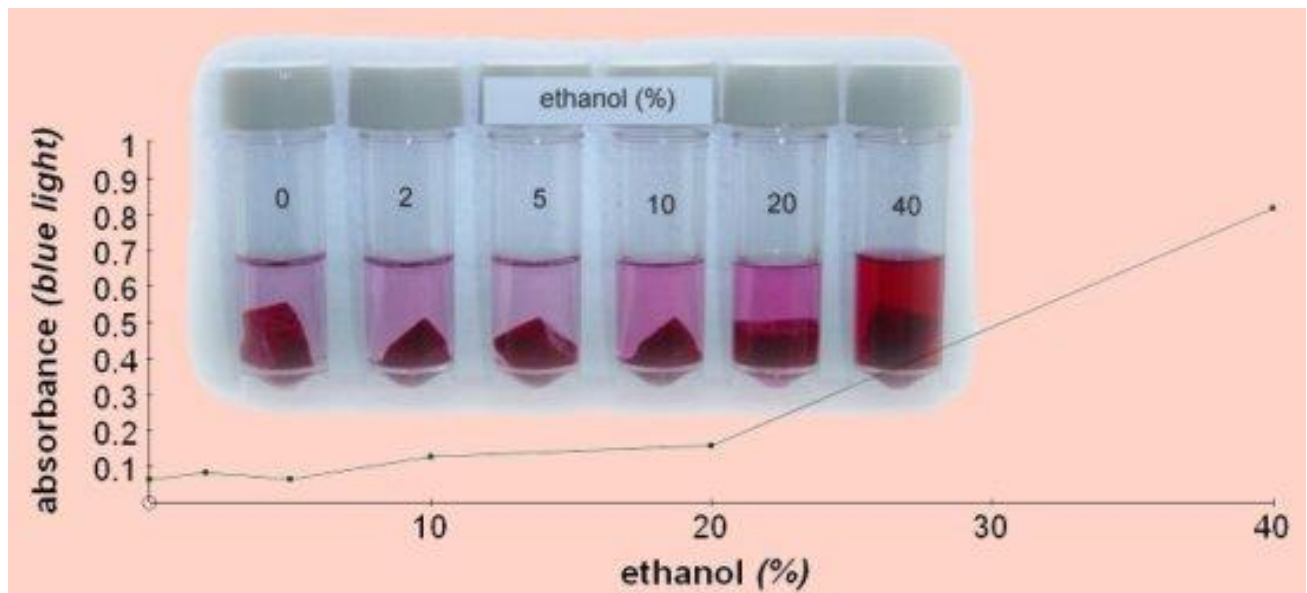


Type of Membrane Protein	Function
Enzymes 酶	Speed up chemical reactions
Channel proteins 通道蛋白	Provide channels for water-soluble molecules to move across the membrane.
Carrier/Transport proteins 輸運蛋白	Transport substances by shape change (needs energy from respiration)
Receptor proteins 受收器蛋白	"Card reader" Bind 結合 specific chemical messengers (e.g., hormones) to regulate cell activities.
Glycoproteins 糖蛋白	"ID Card" Act as antigen (抗原) for cell identification.



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Factors damaging the cell membrane



“Juice leaked 漏汁” = cell membrane damaged!

Factor	Effect on Phospholipids 磷脂	Effect on Membrane Proteins 膜蛋白	Examples
Organic solvents 有機溶劑	Dissolves the lipid bilayer	Denatures 變性 proteins	Ethanol 乙醇, alcohol 酒精
Detergents 清潔劑	Dissolves the lipid bilayer	Break down protein-lipid interactions	Soap (肥皂)
Extreme pH	Destabilizes 不穩定 lipid bilayer	Denatures proteins	Stomach acid
High temperature	Destabilizes the lipid bilayer	Denatures proteins	Boiling
Mechanical stress 機械應力	—	—	Pressure, too much water entering cells

*Applications: kill harmful microorganisms (e.g. bacteria)

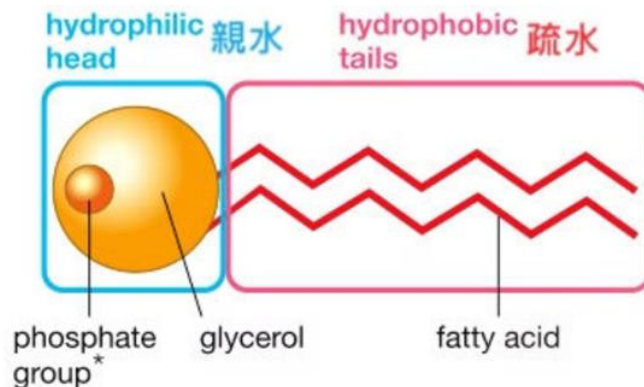
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Concept checks

Multiple Choice Questions (MCQs)

1. Which of the following components of a phospholipid is hydrophilic?

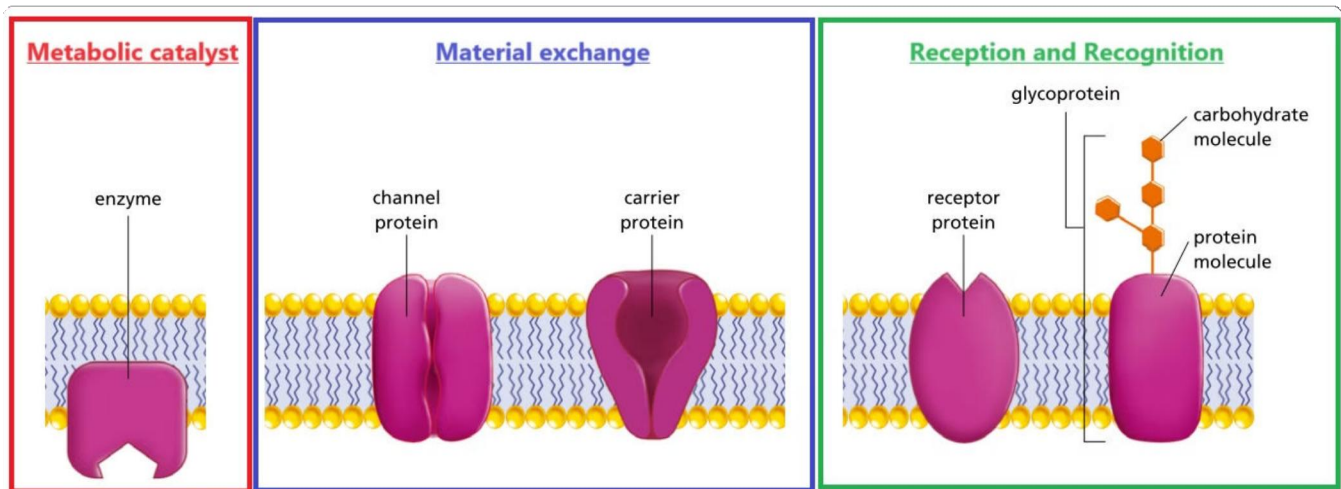
Phospholipid molecule



- A. Fatty acid tails
 - B. Glycerol backbone
 - C. Phosphate group
 - D. Both fatty acid tails and glycerol
2. What is the main reason why phospholipids form a bilayer in water?
- A. Hydrophobic tails are attracted to water.
 - B. Hydrophilic heads interact with water, while hydrophobic tails avoid it.
 - C. Both heads and tails are hydrophilic.
 - D. Phospholipids dissolve completely in water.
3. Which of the following molecules is transported by carrier proteins?
- A. Small, non-polar molecules
 - B. Large, polar molecules
 - C. Water-soluble ions
 - D. Gases like oxygen and carbon dioxide

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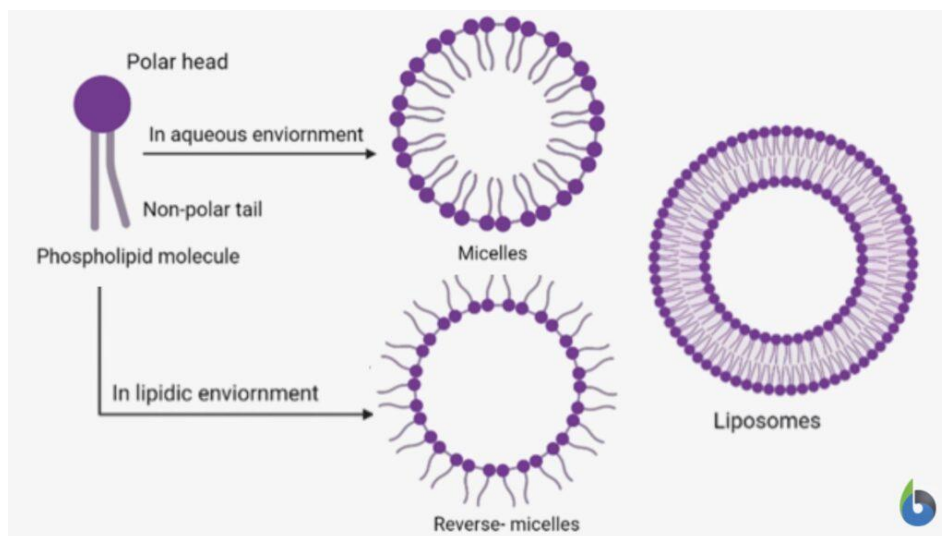
4. Which type of membrane protein acts like an "ID card" for cell identification?



- A. Channel proteins
 - B. Glycoproteins
 - C. Carrier proteins
 - D. Receptor proteins
5. What happens to a protein in cell membrane at high temperatures?
- A. It becomes more rigid.
 - B. It denatures.
 - C. It becomes impermeable.
 - D. It thickens.
6. Which factor can denature membrane proteins but not dissolve the lipid bilayer?
- A. Organic solvents
 - B. Detergents
 - C. Extreme pH
 - D. Adding oil

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7. Which of the following is NOT a function of membrane proteins?
- A. Speed up chemical reactions
 - B. Transport large molecules
 - C. Store genetic material
 - D. Act as receptors for chemical messengers
8. What is the role of enzymes 酶 in the cell membrane?
- A. To provide structural support
 - B. To speed up chemical reactions
 - C. To transport molecules across the membrane
 - D. To regulate cell recognition
9. Why is the phospholipid bilayer described as "tail-to-tail"?



- A. Hydrophilic 親水 heads face each other.
- B. Hydrophobic 疏水 tails face each other inside the bilayer.
- C. Hydrophilic heads face the cytoplasm only.
- D. Hydrophobic tails face the extracellular 細胞外 fluid.

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10. Which of the following factors can dissolve 溶解 the lipid bilayer?

- A. Mechanical stress
- B. Detergents
- C. Extreme pH
- D. High temperature

True/False Questions

Directions: Put a "✓" if the sentence is correct and put a "✗" on the phrase or words that are incorrect.

1. The phosphate group in a phospholipid is hydrophobic.
2. Membrane proteins are embedded in a mosaic pattern in the bilayer.
3. Organic solvents dissolve the lipid bilayer and denature proteins.
4. Channel proteins transport substances by changing shape.
5. Glycoproteins in the membrane act as antigens for cell identification.

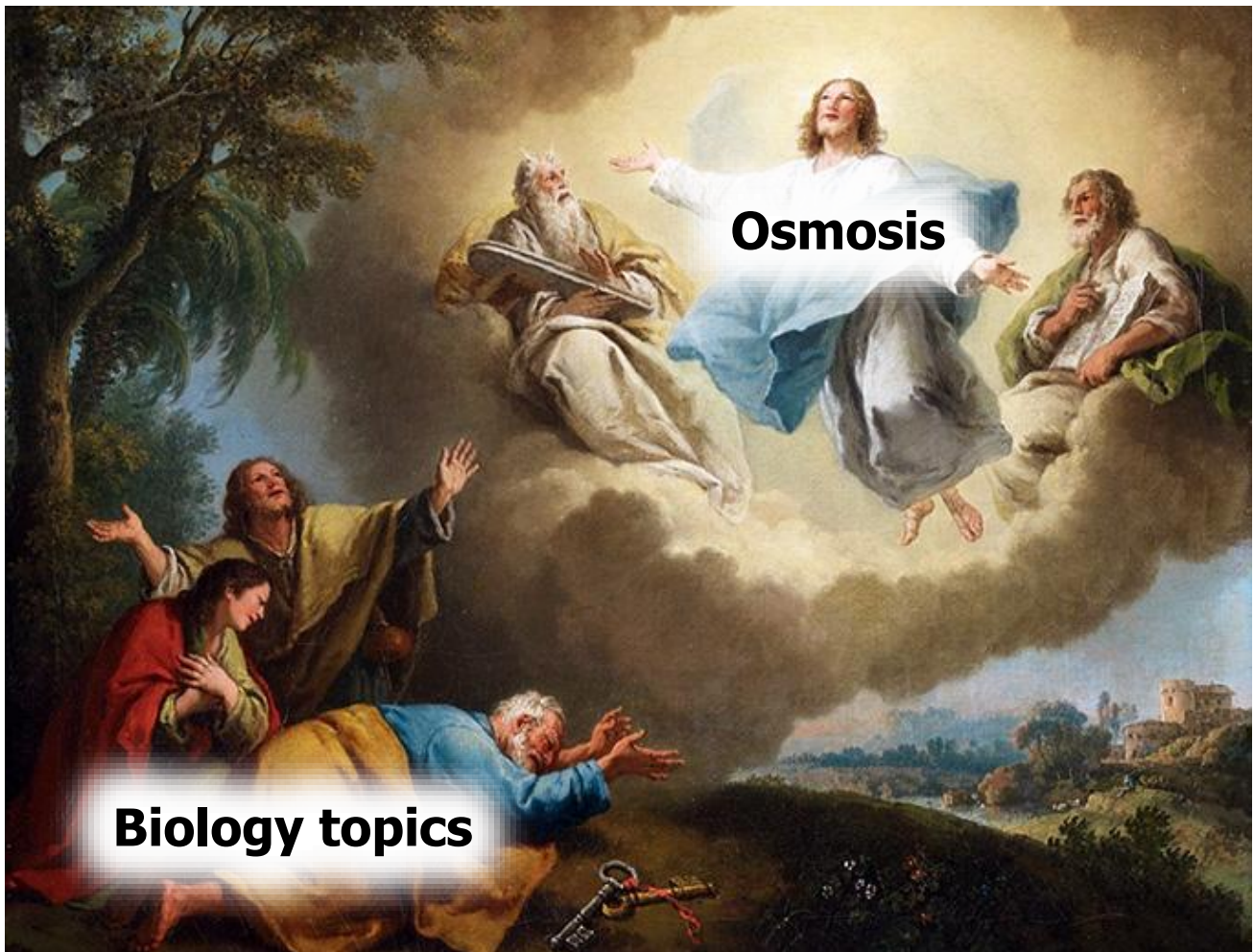
Fill in the Blanks

☐ enzymes 酶 ☐ receptor proteins 受體蛋白 ☐ hydrophobic 疏水 ☐ glycoproteins 糖蛋白
☐ phospholipid bilayer 磷脂雙層 ☐ detergents 清潔劑 ☐ hydrophilic 親水

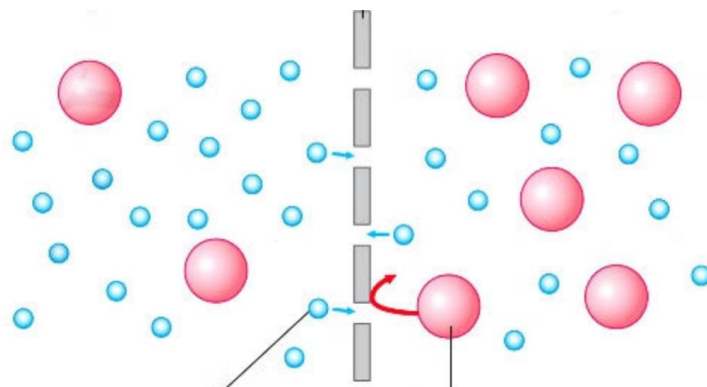
1. The _____ forms the basic structure of the cell membrane.
2. The fatty acid tails of phospholipids are _____.
3. The phosphate group in a phospholipid is _____.
4. _____ speed up chemical reactions in the membrane.
5. _____ bind to specific chemical messengers to regulate cell activities.
6. _____ act as "ID cards" for cell identification.
7. _____ can dissolve the lipid bilayer by breaking protein-lipid interactions.

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Osmosis 滲透作用 – net movement of water molecules



Concept of net 淨



Movement	Net movement
► 10 molecules	► 10
◄ 10 molecules	
► 5 molecules	
◄ 5 molecules	

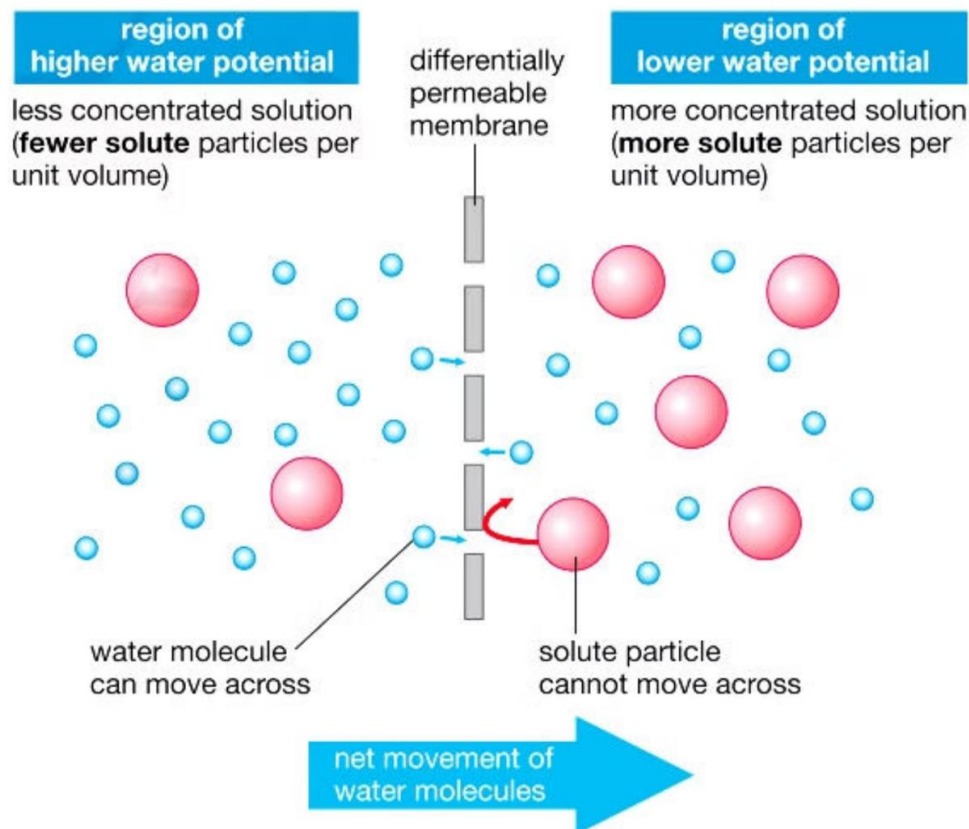
No net movement ≠ no movement!

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Requirements of osmosis

1. Differentially permeable 差異滲透性 membrane
2. Water potential difference 水勢

Solution	Water potential (kPa)
Pure water	0
5% sucrose solution	-2
10% sucrose solution	-10



Less solute (higher water concentration)	More solute (lower water concentration)
Higher water potential	Lower water potential
► Net water movement ►	

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In other words: Water follows solute! 水跟溶質走

(in presence of differentially permeable membrane)

Osmosis is

the **net movement of water molecules**

across a **differentially permeable membrane**

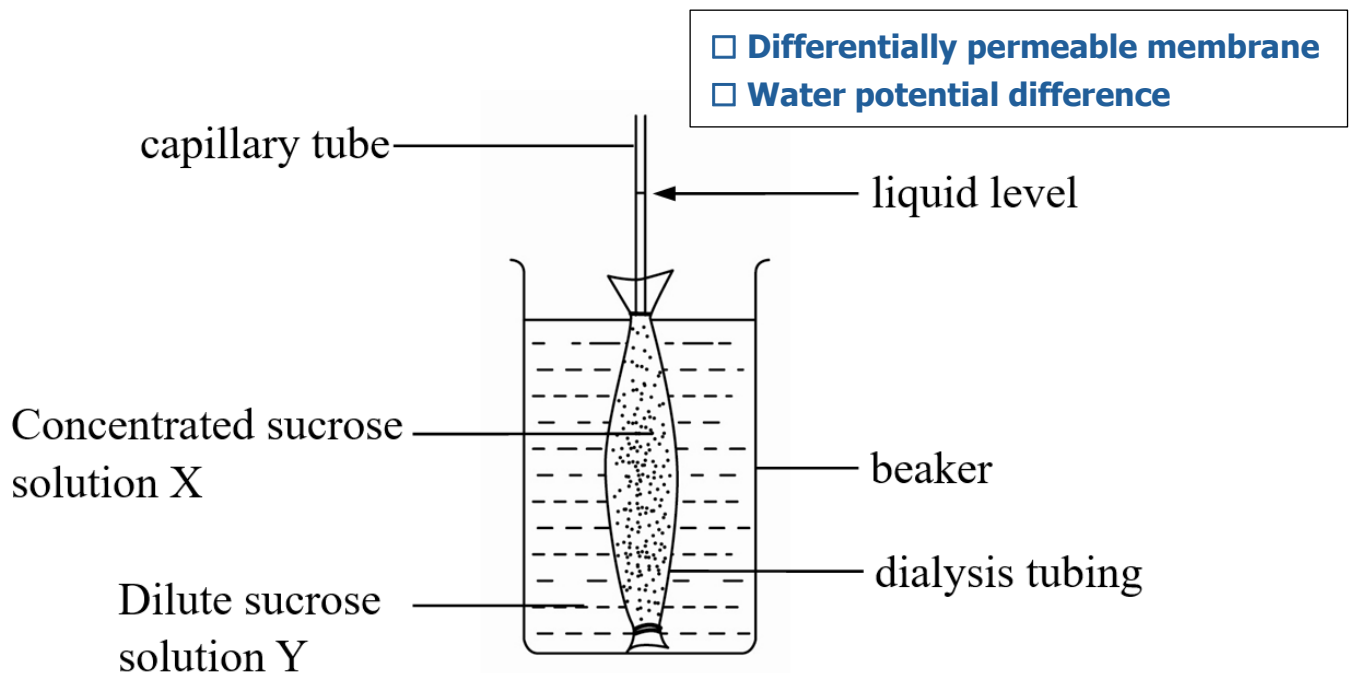
from a region of **higher water potential to a region of lower water potential.**

水分子穿過差異滲透膜，從水勢較高的區域向水勢較低的區域的淨移動

When there is no water potential (solute concentration) difference between two regions, osmosis stops. (but remember no net movement $\neq$ no movement!)

Osmosis **does not require energy.** It is a **passive** 被動的 process.

The following set-up was used to investigate osmosis.

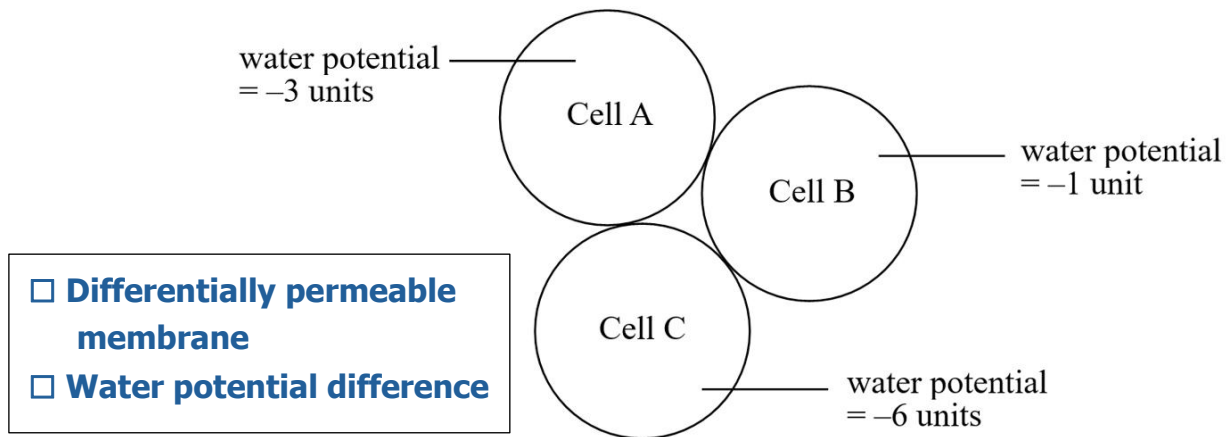


30 minutes after the beginning of the experiment, the liquid level in the glass tube had risen because

- A. water molecules pass from X to Y.
- B. water molecules pass from Y to X.
- C. more water molecules pass from X to Y than from Y to X.
- D. more water molecules pass from Y to X than from X to Y.

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The diagram below shows three cells (A, B and C) which are in contact with each other. The water potential of the three cells is different. Draw arrows in the diagram to show the direction of the net movement of water molecules among these cells. (3 marks)



Setup A

Distilled water
Beet root cylinder



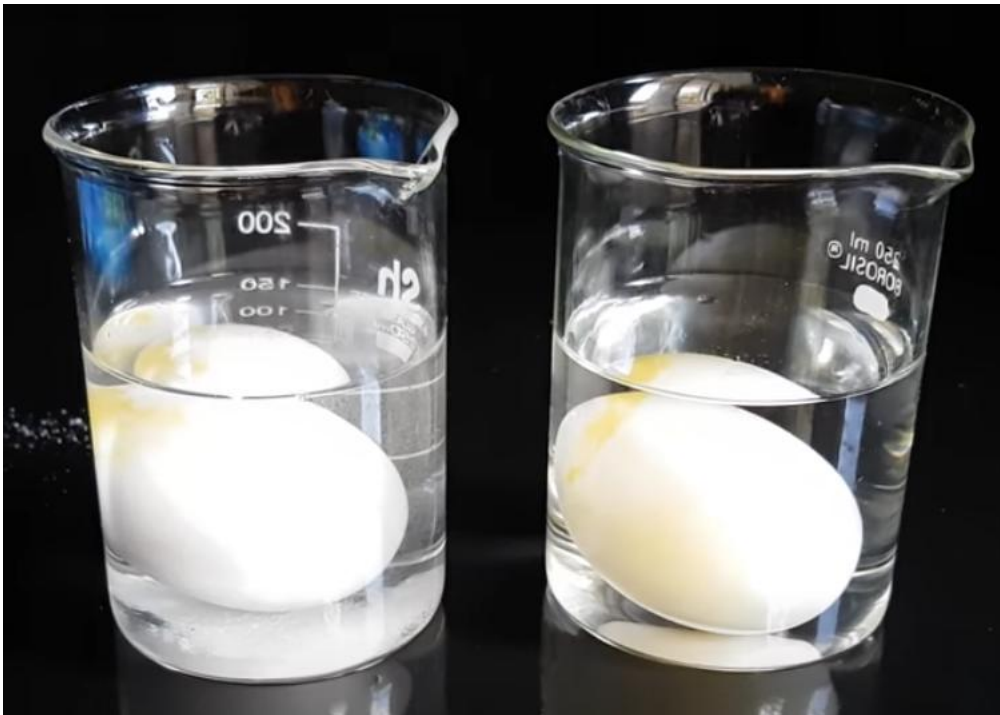
Setup B



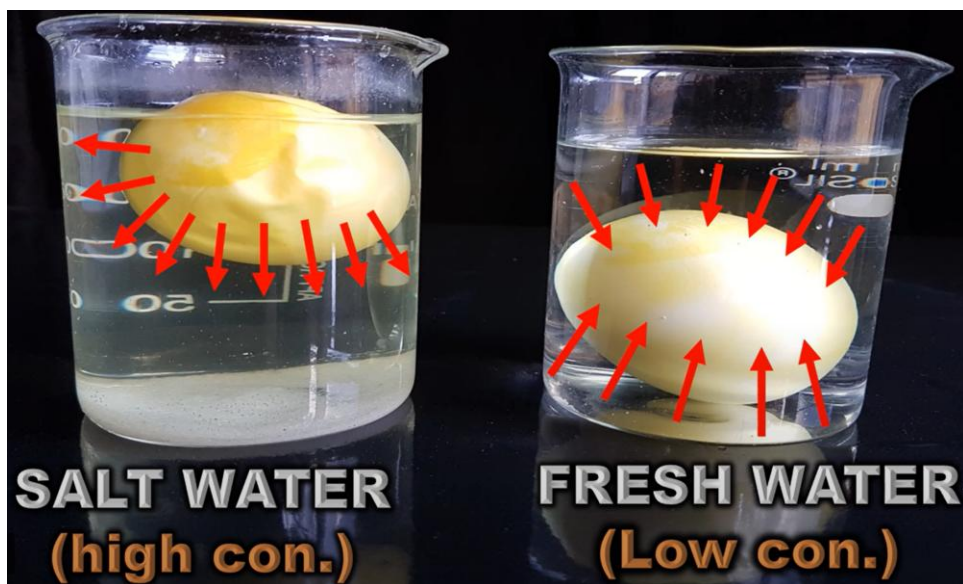
Water potential of distilled water	+ve / 0 / -ve
Water potential of beet root cells	+ve / 0 / -ve

	Setup A	Setup B
Juice leaked?	Yes / No	Yes / No
	☐ Differentially permeable membrane ☐ Water potential difference	☐ Differentially permeable membrane ☐ Water potential difference
Osmosis?	Yes / No	Yes / No
Net water movement	Distilled water (◀/▶) beet root	
Cylinder would become	Longer / shorter Heavier / lighter Denser / less dense	

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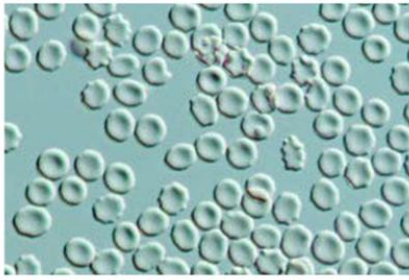
	In concentrated salt solution	In tap water
Osmosis?	☐ Differentially permeable membrane ☐ Water potential difference	☐ Differentially permeable membrane ☐ Water potential difference
Water potential	Higher / lower than the egg	Higher / lower than the egg
Net water movement	Into / out of the egg	Into / out of the egg
Egg	Expand / Shrink	Expand / Shrink



Why do we need to remove the egg shell before the experiment?

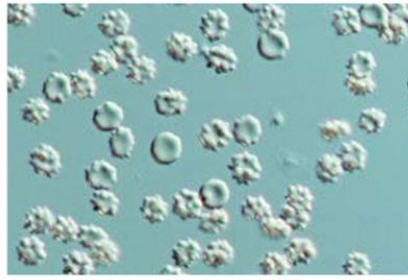
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The following photomicrographs show the appearance of red blood cells immersed in different concentrations of sucrose solution (A, B and C).

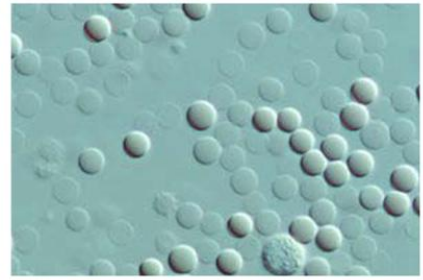


sucrose solution A

Normal appearance



sucrose solution B



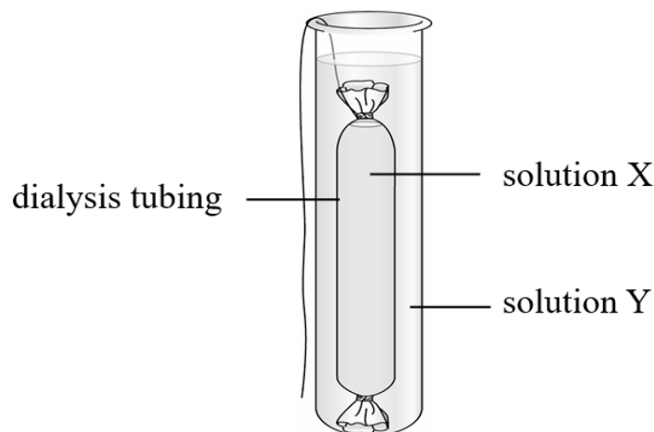
sucrose solution C

Which of the following correctly arranges the concentrations of sucrose solutions A, B and C in descending order?

- A. $A > C > B$
- B. $B > A > C$
- C. $C > A > B$
- D. $C > B > A$

An experiment was carried out to investigate osmosis using the following set-up.

The initial mass and final mass of the tubing was measured as shown in the table below.



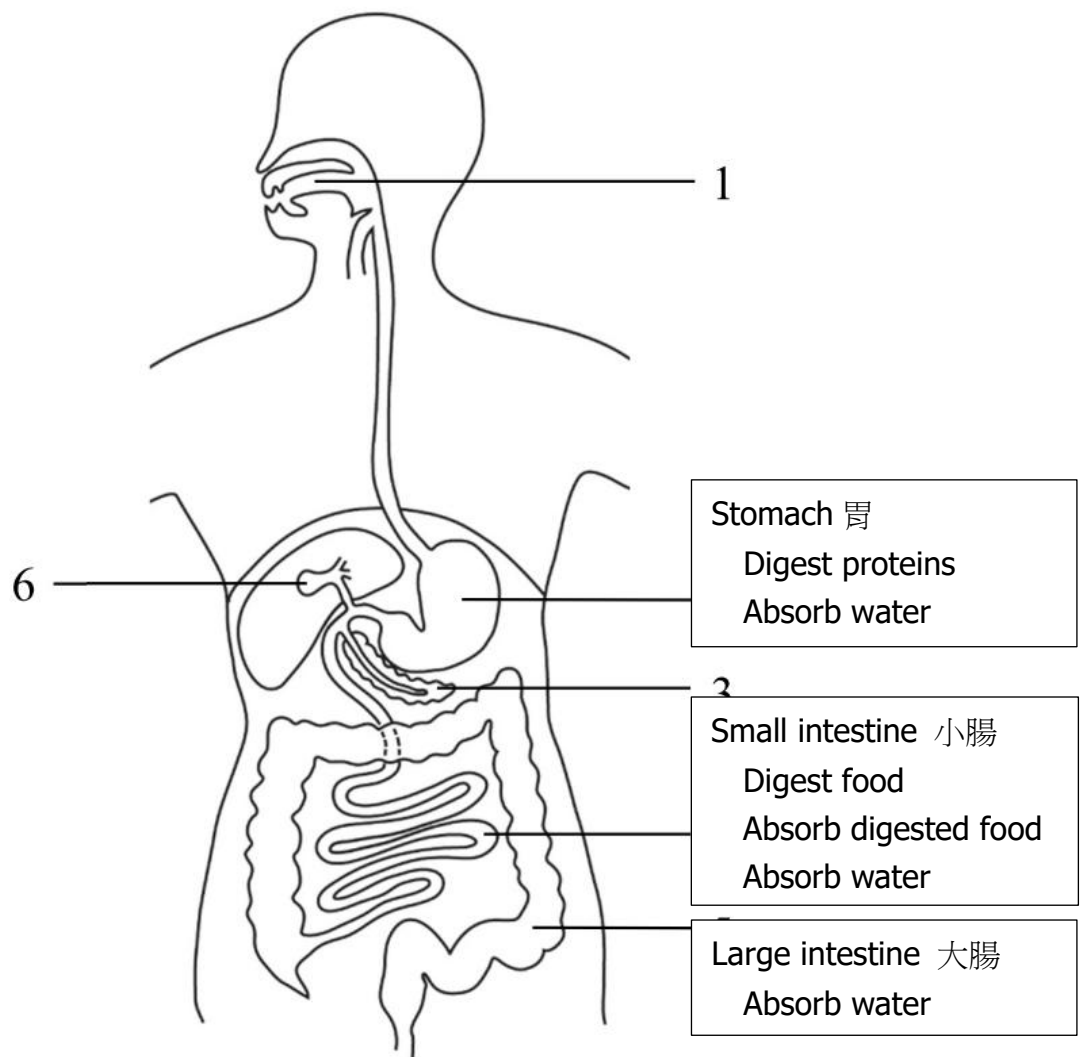
Initial mass	12.0 g
Final mass	9.5 g

These results would be obtained when

- A. X is a 5% sucrose solution and Y is distilled water.
- B. X is a 5% sucrose solution and Y is a 10% sucrose solution.
- C. X is a 10% sucrose solution and Y is a 5% sucrose solution.
- D. X is a 10% sucrose solution and Y is distilled water.

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Water follows solute

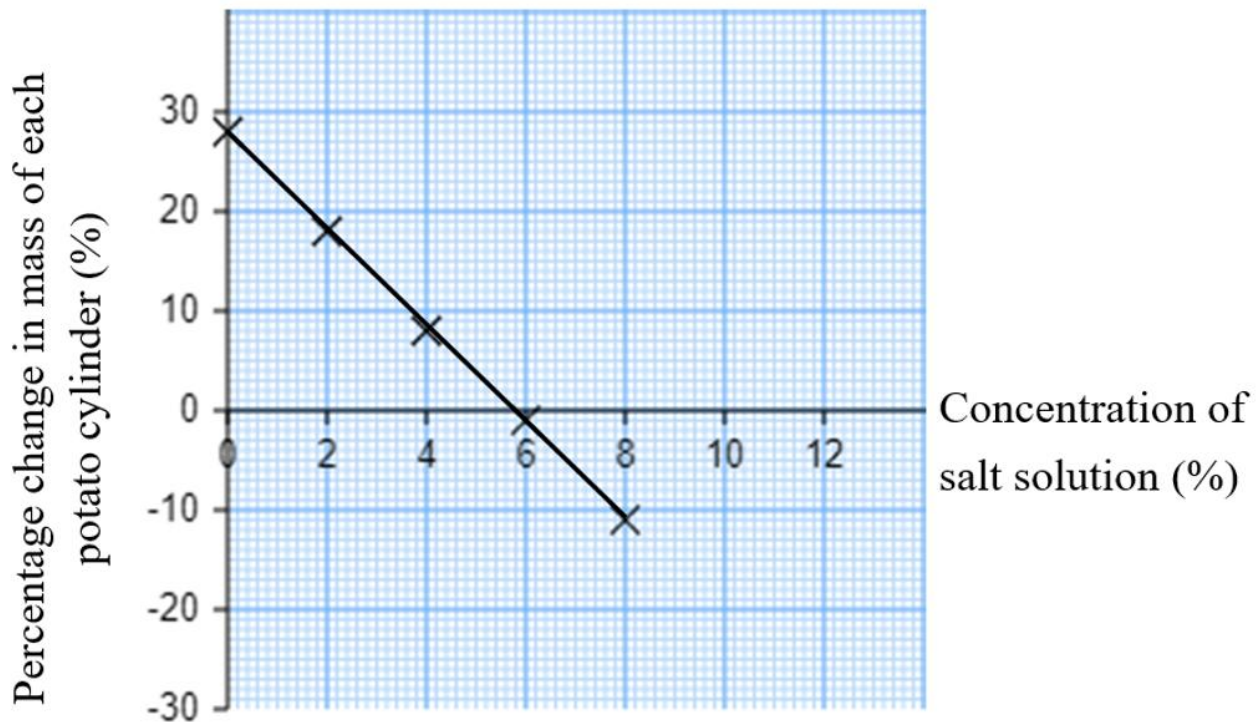


Which part of the human digestive system absorbs the most amount of water?

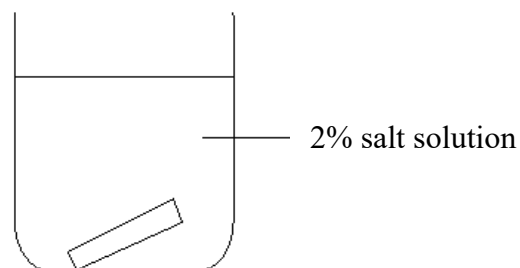
- A. Stomach
- B. Small intestine
- C. Large intestine
- D. All parts absorb the same amount of water

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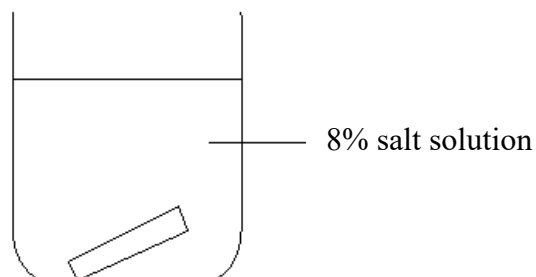
A student carried out an investigation to determine the water potential of potato tissue. She cut 5 cylinders from a potato. She measured the mass of each cylinder and then placed each one in salt solution (0%, 2%, 4%, 6% and 8%). After 1 hour, she measured the mass of each cylinder again and calculated the percentage change in mass. The results were plotted in the graph below.



The water potential of the potato cylinder is [higher / lower / same] as 2% salt solution



The water potential of the potato cylinder is [higher / lower / same] as 8% salt solution

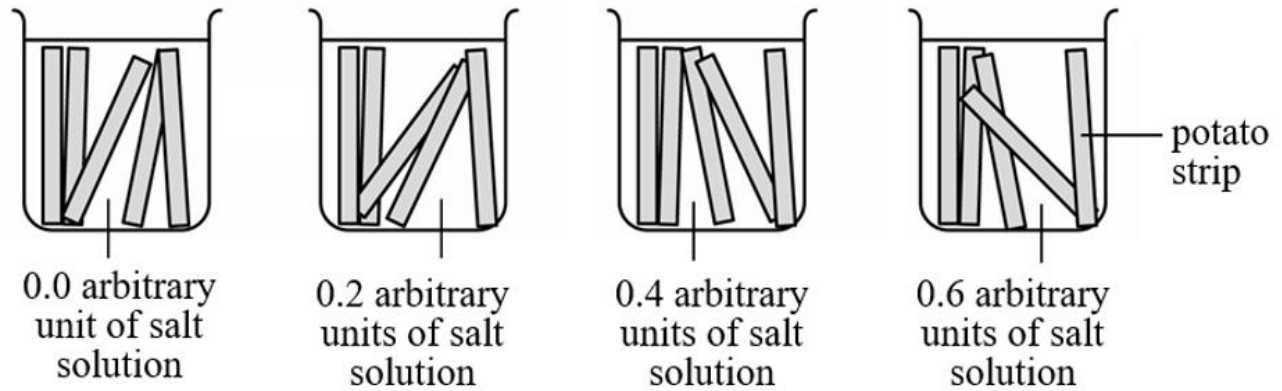


The water potential of the potato cylinder is the same as _____% salt solution

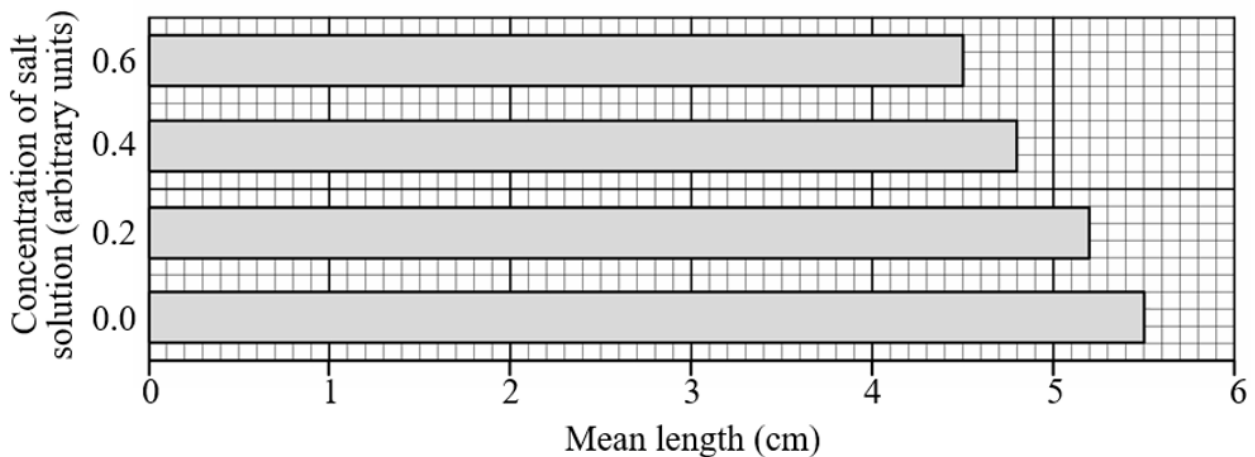
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In the experiment, several thin strips with equal size were prepared from a peeled potato. All of them were **5 cm** long.

They were placed in salt solutions of different concentrations as shown below.



After two hours, the potato strips were taken out from the solutions. Their lengths were measured, and the results are shown in the following chart.



In which concentration of salt solution would you expect the length of the potato strip to remain unchanged?

- A. 0.0 arbitrary units
- B. 0.1 arbitrary units
- C. 0.3 arbitrary units
- D. 0.6 arbitrary units

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Concept checks

Multiple Choice Questions (MCQs)

1. What is the primary requirement for osmosis to occur?
 - A. Light
 - B. Temperature
 - C. Differentially permeable membrane
 - D. High solute concentration

2. Which solution has the highest water potential?
 - A. Pure water
 - B. 5% sucrose solution
 - C. 10% sucrose solution
 - D. Saltwater

3. In the context of osmosis, what does a lower water potential indicate?
 - A. Higher solute concentration
 - B. Lower solute concentration
 - C. Equal solute concentration
 - D. No solute

4. During osmosis, water moves from:
 - A. High solute concentration to low solute concentration
 - B. Low water potential to high water potential
 - C. High water potential to low water potential
 - D. Low solute concentration to high solute concentration

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5. What is the role of a differentially permeable membrane in osmosis?
 - A. It allows solute movement
 - B. It blocks all substances
 - C. It allows only water molecules to pass through
 - D. It facilitates active transport

6. Which of the following is true about osmosis?
 - A. It requires energy
 - B. It is an active process
 - C. It is a passive process
 - D. It involves the movement of solutes

7. If a beetroot cell leaks pigment, will osmosis still occur?
 - A. Yes, because the membrane is still partially permeable.
 - B. No, because the membrane is damaged.
 - C. Yes, because pigment leakage does not affect water movement.
 - D. No, because osmosis requires pigment retention.

8. Which of the following statements is correct regarding water potential?
 - A. It increases with solute concentration
 - B. It decreases with solute concentration
 - C. It remains constant regardless of solute concentration
 - D. It is not affected by solute concentration

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9. If a potato is boiled, will osmosis still occur?
- A. Yes, because the cell wall remains intact.
 - B. No, because the cell membrane is damaged.
 - C. Yes, because boiling does not affect osmosis.
 - D. No, because boiling increases the water potential.
10. Which condition will stop osmosis?
- A. Equal solute concentration on both sides
 - B. High temperature
 - C. Low temperature
 - D. Increased light intensity
11. What is the water potential of pure water?
- A. -10 kPa
 - B. -2 kPa
 - C. 0 kPa
 - D. +10 kPa
12. Which of the following is not a factor affecting osmosis?
- A. Temperature
 - B. Solute concentration
 - C. Light intensity
 - D. Water potential

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13. In the presence of a differentially permeable membrane, osmosis will occur when:

- A. Solute concentration is equal
- B. Water potential is equal
- C. There is a water potential difference
- D. There is no solute present

14. How does the presence of a solute affect water potential?

- A. It increases water potential
- B. It decreases water potential
- C. It has no effect on water potential
- D. It doubles the water potential

15. Which solution has the lowest water potential?

- A. Pure water
- B. 5% sucrose solution
- C. 10% sucrose solution
- D. Distilled water

16. What does a negative water potential indicate?

- A. Absence of solute
- B. Presence of solute
- C. High water concentration
- D. Low water concentration

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17. Which statement is true about the direction of net water movement?

- A. It moves from low to high solute concentration
- B. It moves from high to low solute concentration
- C. It moves from low to high water potential
- D. It moves from high to low water potential

18. Which factor is NOT required for osmosis?

- A. Differentially permeable membrane
- B. Water potential difference
- C. Solute movement
- D. Presence of water

19. What is the immediate effect of placing a cell in a solution with lower water potential?

- A. Cell shrinks
- B. Cell swells
- C. No change in cell size
- D. Cell bursts

20. When two solutions are separated by a differentially permeable membrane, water moves towards:

- A. Higher water potential
- B. Lower water potential
- C. Higher solute concentration
- D. Lower solute concentration

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True/False Questions

Direction: put a "✓" if the sentence is correct and put a "X" on the phrase or words that are wrong.

1. Osmosis requires energy input.
2. Water moves from regions of high to low water potential during osmosis.
3. A differentially permeable membrane is necessary for osmosis.
4. Solute particles move across the membrane during osmosis.
5. Osmosis is considered a passive process.
6. A region with higher solute concentration has higher water potential.
7. Water potential is higher in pure water than in a sucrose solution.
8. Osmosis can occur without a water potential difference.
9. The presence of solute decreases water potential.
10. Water moves from dilute to concentrated solutions in osmosis.
11. In osmosis, water moves towards the region with lower solute concentration.
12. The net movement of water in osmosis is from low to high solute concentration.
13. Osmosis stops when there is no water potential difference.
14. A higher solute concentration results in a more negative water potential.
15. Osmosis occurs only in plant cells.
16. The presence of a solute increases the water potential.
17. Water potential is equal in all solutions.
18. Osmosis cannot occur without a membrane.
19. Solute concentration does not affect osmosis.
20. Osmosis involves the movement of solute particles.

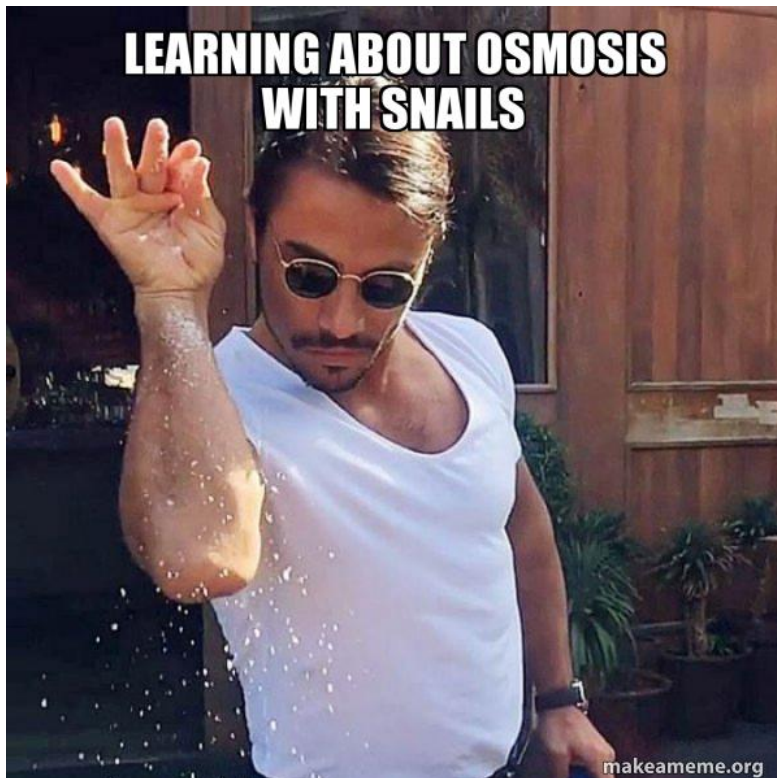
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Fill in the Blanks

Choose from the following:

- ☐ solute concentration 溶質濃度 ☐ passive 被動 ☐ differentially permeable 差異滲透性
☐ water potential 水勢 ☐ osmosis 滲透作用

1. _____ is the process where water moves across a differentially permeable membrane.
2. The _____ difference drives the movement of water in osmosis.
3. Osmosis is a _____ process and does not require energy.
4. A _____ membrane is necessary for osmosis to occur.
5. In osmosis, water moves from a region of low _____ to high _____.



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Answer Key

MCQs

1. C
2. A
3. A
4. C
5. C
6. C
7. B
8. B
9. B
10. A
11. C
12. C
13. C
14. B
15. C
16. B
17. D
18. C
19. A
20. B

True/False

1. X (Osmosis does not require energy)
2. ✓
3. ✓
4. X (Solute particles do not move across the membrane)
5. ✓
6. X (Higher solute concentration means lower water potential)
7. ✓
8. X (Osmosis requires a water potential difference)
9. ✓
10. ✓

11. X (Water moves towards higher solute concentration)
12. X (Water moves from low to high water potential)
13. ✓
14. ✓
15. X (Osmosis occurs in both plant and animal cells)
16. X (Presence of solute decreases water potential)
17. X (Water potential varies with solute concentration)
18. ✓
19. X (Solute concentration affects osmosis)
20. X (Osmosis involves the movement of water)

Fill in the Blanks

1. Osmosis 滲透作用
2. Water potential 水勢
3. Passive 被動
4. Membrane 膜
5. Solute concentration 溶質濃度 / Water potential 水勢

Answer Key

Concept check – fluid mosaic model

MCQs

1. B
2. C
3. B
4. B
5. B
6. D
7. B
8. C
9. C
10. B

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True/False

1. ✗ (Replace "rigid structure" with "flexible structure")
2. ✓
3. ✗ (Replace "small, non-polar molecules" with "large or charged molecules")
4. ✓
5. ✗ (Replace "without assistance" with "with assistance from proteins")

Fill in the Blanks

1. phospholipid bilayer 磷脂雙層
2. hydrophilic 親水性
3. hydrophobic 疏水性
4. proteins 蛋白質
5. selective permeability 差異滲透性

Concept check – phospholipid, protein and cell membrane stability

Answer Key

MCQs

1. C
2. B
3. B
4. B
5. B
6. C
7. C
8. B
9. B
10. B

True/False

1. ✗ (Replace "hydrophobic" with "hydrophilic")
2. ✓
3. ✓
4. ✗ (Replace "changing shape" with "providing channels")

5. ✓

Fill in the Blanks

1. phospholipid bilayer 磷脂雙層
2. hydrophobic 疏水
3. hydrophilic 親水
4. enzymes 酶
5. receptor proteins 受體蛋白
6. glycoproteins 糖蛋白
7. detergents 清潔劑

Concept check – osmosis

Answer Key

MCQs

1. C
2. A
3. A
4. C
5. C
6. C
7. B
8. B
9. B
10. A
11. C
12. C
13. C
14. B
15. C
16. B
17. D
18. C
19. A
20. B

True/False

1. ✗ (Osmosis does not require energy)
2. ✓
3. ✓

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4. ✗ (Solute particles do not move across the membrane)
5. ✓
6. ✗ (Higher solute concentration means lower water potential)
7. ✓
8. ✗ (Osmosis requires a water potential difference)
9. ✓
10. ✓
11. ✗ (Water moves towards higher solute concentration)
12. ✗ (Water moves from low to high water potential)
13. ✓
14. ✓
15. ✗ (Osmosis occurs in both plant and animal cells)

16. ✗ (Presence of solute decreases water potential)
17. ✗ (Water potential varies with solute concentration)
18. ✓
19. ✗ (Solute concentration affects osmosis)
20. ✗ (Osmosis involves the movement of water)

Fill in the Blanks

1. Osmosis 滲透作用
2. Water potential 水勢
3. Passive 被動
4. Membrane 膜
5. Solute concentration 溶質濃度 /
Water potential 水勢